JK. Class-1

Series Solution

Analytic function: A function f is said to be analytic at x_0 if it's taylor seires about x

$$\sum_{n=0}^{\infty} \frac{f^n(x_0)}{n!}(x-x_0)^n$$ exists and converage

Example: $f(x) = x^5 + 3x^4 + 2x^2 + 8$ is analytic everywhere

$f(x) = \frac{1}{x^2 - 3x + 2}$ analytic for all points except $x = 1$ and 2

Singular Point:

Diff. equ.

Consider the $\underset{\sim}{DE}$ $a_0(x)y'' + a_1(x)y' + a_2(x)y$

re-write the DE in the equivalent normalized from,

$$y'' + \frac{a_1(x)}{a_0(x)}y' + \frac{a_2(x)}{a_0(x)}y = 0$$

$\Rightarrow$ $y'' + P_1(x)y' + P_2(x)y = 0$ — (1) where $P_1(x) = \frac{a_1(x)}{a_0(x)}$, $P_2(x) = \frac{a_2(x)}{a_0(x)}$

If at least one of the functions $P_2(x)$ and $P_1(x)$ is not analytic at x_0, then x_0 is called a singular point.

Let, $P_1(x) = -1/2x$ and $P_2(x) = \frac{x-5}{2x^2}$

Since both $P_1(x)$ & $P_2(x)$ are not analytic at $x=0$, so $x=0$ is a singular point.

Regular Singular Point:

Let x_0 is a singular Point of the DE (1). If the function defined by the products $(x-x_0)P_1(x)$ and $(x-x_0)^2 P_2(x)$ are both analytic at x, then x_0 is called a regular singular point of the given DE.

Irregular Singular Point:

If either (or both) of the functions defined by the products $(x-x_0)P_1(x)$ and $(x-x_0)^2 P_2(x)$ is not analytic at x_0 then x_0 is called irregular singular Point of the given DE (1).

Example:

Consider $y'' - \frac{1}{2x} y' + \frac{x-5}{2x^2} y = 0$

Here,

$P_2(x) = \frac{-1}{2x}$, $P_1(x) = \frac{x-5}{2x^2}$

Since both $P_1(x)$ & $P_2(x)$ fail to be analytic at $x=0$

So, $x=0$ is a singular point of

Now, consider the function defined by $x P_2'(x) = \frac{-1}{2}$ and $x^2 P_2(x) = \frac{x-5}{2}$. Since both of these product function are analytic at $x=0$. So $x=0$ is a regular singular point.

Example:

Let the DE, $y'' + \frac{2}{x^2(x-2)} y' + \frac{x+1}{x^2(x-2)^2} y = 0$

Sol^n

Here, $P_1(x) = \frac{2}{x^2(x-2)}$, $P_2(x) = \frac{x+1}{x^2(x-2)^2}$

Here $x=0$ and $x=2$ are singular point

Test $x=0$

$x P_1(x) = \frac{2}{x(x-2)}$ and $x^2 P_2(x) = \frac{x+1}{(x-2)^2}$

Since $x P_1(x)$ is not analytic at $x=0$

but $x^2 P_2(x)$ is analytic at $x=0$

So, $x=0$ is a irregular singular point.

Test $x=2$

$(x-2) P_1(x) = 2/x^2$ and

$(x-2)^2 P_2(x) = \frac{x+1}{x^2}$

$\therefore$ Both ~~are~~ of the product analytic ~~are~~

at $x = 2$ so, regular singular Point.